

QUD-sensitive projection in a dynamic semantics with alternatives

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Abstract. This paper presents a compositional dynamic system (building on Muskens 1996; Stone 1999; Brasoveanu 2006; Farkas and Bruce 2010; AnderBois et al. 2015), in which question alternatives are introduced via the random assignment of propositional discourse referents. This system provides a framework for implementing a mechanism for presupposition projection that is sensitive to the question under discussion (QUD; Ginzburg 1996; Roberts 2012; Büring 2003; Farkas and Bruce 2010). The analysis addresses the triggering problem by treating presuppositions as entailments (Abrusán 2011) and derives projection for not-at-issue entailments, following work linking projection to at-issueness (e.g., Simons 2001; Simons et al. 2017; Tonhauser et al. 2018; Degen and Tonhauser 2025). Projection is derived by computing similarity to local context sets for over QUD-induced partitions of the context set, yielding a compositional account of QUD-sensitive projection.

Keywords: dynamic semantics · QUD · projection.

1 Introduction

Presupposition projection is sensitive to the discourse context: the same sentence may give rise to a projective inference in one context but not in another. This paper develops a formal account of this context sensitivity, focusing on the role of the question under discussion (QUD; Ginzburg 1996; Roberts 2012; Büring 2003; Farkas and Bruce 2010).

To illustrate the phenomenon of projection, consider (1a), where the modal preajacent *Anna quit smoking* entails that Anna does not currently smoke (1b). In addition, it gives rise to the inference that Anna used to smoke (1c), which may persist under entailment-cancelling operators, i.e., it may project.

- | | | | |
|-----|----|---------------------------------------|-------------------------------|
| (1) | a. | <i>Maybe Anna quit smoking.</i> | $(\Diamond(u \wedge \neg s))$ |
| | b. | <i>Anna does not currently smoke.</i> | $(\neg s)$ |
| | c. | <i>Anna used to smoke.</i> | (u) |

A central empirical generalization is that projection can depend on the current QUD, as suggested by a body of work linking projection to at-issueness, the QUD, and information structure (e.g., Abbott 2000; Simons 2001; Qing et al. 2016; Simons et al. 2017; Tonhauser et al. 2018; Degen and Tonhauser 2025). The

- (2) a. Q: *Does Anna smoke?* (?S)
 b. A: *Maybe she quit smoking.* $(\Diamond(\mathbf{u} \wedge \neg s))$
 c. Suggests answer: Maybe A doesn't smoke $(\Diamond\neg s)$
 d. Projects "Anna used to smoke" (u)

In contrast, no such projective inference arises in the example in (3) (based on [Simons 2001](#): p. 432, attributed to [Geurts 1994](#)). (3a) raises the question of why Anna is nervously chewing her pencil, and the response (3b) is interpreted as addressing a sub-question about whether Anna quit smoking which could be the reason for Anna's chewing her pencil (see e.g., discussion in [Simons 2001](#); [Qing et al. 2016](#)). In this context, the modal assertion therefore suggests that the answer to the question may be $u \wedge \neg s$. In this context, no inference about whether or not Anna smoked in the past is made, and thus no projection arises.

- (3) a. Q: *Why is Anna nervously chewing her pencil?* ($?(\mathbf{u} \wedge \neg\mathbf{s})$)
 b. A: *Maybe she quit smoking.* ($\Diamond(\mathbf{u} \wedge \neg\mathbf{s})$)
 c. Suggests answer: Maybe A quit smoking
 (and that's why A is chewing her pencil) ($\Diamond(\mathbf{u} \wedge \neg\mathbf{s})$)
 d. No projection

The contrast between (2) and (3) shows that projection may depend on how an utterance relates to the QUD. However, no existing formal semantic account derives this QUD-sensitivity. While there are theoretical advances (Abrusán 2011; Simons et al. 2010; Roberts and Simons 2024) and explicit pragmatic approaches in the RSA framework (Qing et al. 2016; Warstadt 2022; Scontras and Tonhauser 2025), no formal semantic analysis has been made explicit.

This paper develops such an account within a compositional dynamic framework. Building on Plural Compositional DRT (PCDRT; [Muskens 1996](#); [Brasoveanu 2006, 2010](#)), the analysis integrates components of the Table model ([Farkas and Bruce 2010](#)). In particular, the system uses propositional variables to model the current context set (following [AnderBois et al. 2010, 2015](#); [Hofmann 2022, 2025](#)), and QUD-alternatives are modeled via the random introduction of propositional discourse referents that introduce partitions over the context set.

Propositional variables also store local contexts introduced by propositional operators (Stone 1999; Stone and Hardt 1999; Brasoveanu 2006, 2010; AnderBois et al. 2010, 2015; Snider 2017; Hofmann 2019, 2022, 2025), and QUD-sensitive projection is derived by pruning QUD-alternatives by choosing the worlds most *similar* to the local context. The similarity-based mechanism relating global and local contexts is inspired by similarity-based analyses of counterfactual conditionals (Stalnaker 1968; Lewis 1973), and can be viewed as formalization of a

general version of Karttunen’s default principle for belief states invoked in attitude reports (Karttunen 1974; Heim 1992; Sharvit 1998; Sudo 2014).

To derive QUD-sensitive projection, similarity to the local context is evaluated relative to QUD-induced partitions of the context set. This derives projection as a form of global accommodation constrained by the QUD.

The structure of the paper is as follows. Section 2 introduces the intuition behind deriving projection as similarity over context sets and the notion of similarity. Section 3 introduces the formal system and its treatment of context sets and QUD-alternatives. Section 4 illustrates the derivation of the contrast between (2) and (3). Section 5 concludes.

2 Projection as similarity over context sets

Analyses of counterfactual conditionals (following Lewis 1973) derive their truth conditions by requiring local contexts to be similar to the global context, modulo the truth of the embedded content. The present proposal reverses this perspective: projection is derived by requiring the global context to be similar to local contexts introduced by embedded material.

This idea can be understood as a generalization of Karttunen’s default assumption about belief similarity between information states in a discourse (Karttunen 1974; Heim 1992; Sharvit 1998; Sudo 2014), stated as (4):

- (4) *Karttunen’s default principle (for attitude verbs):*
In the absence of evidence to the contrary, assume that the attitude holder shares beliefs with the conversational participants.

While this principle has been discussed primarily for attitude reports, it can be generalized to any kind of hypothetical and counterfactual local contexts:

- (5) *Generalized default principle:* In the absence of evidence to the contrary, assume that the (global and local) context sets invoked in a discourse are similar to one another, i.e., what holds in the global context also holds in local contexts, and vice versa.

The present analysis derives projection from a specific version of this assumption, stated as (6).

- (6) *Default projection principle:*
In the absence of evidence to the contrary, assume that the global context is similar to the local context.

The projection mechanism implemented here can therefore be understood as an instance of a more general pragmatic mechanism, using a default assumption of similarity between context sets, unless explicitly contradicted.

2.1 Sentence meanings and model universe

Following Abrusán 2011, the presupposition of *Maybe Anna quit smoking* is

treated as an entailment of the modal preajcent: *Anna quit smoking* denotes $\{w_u\}$, the set of worlds where Anna used smoke but no longer does ($u \wedge \neg s$). For illustration, assume the model given in (7) and illustrated in Fig. 1.

- (7) a. $D_w = \{w_{us}, w_s, w_u, w_0\}$
 b. A smokes: $I(s) = \{w_{us}, w_s\}$
 c. A used to smoke: $I(u) = \{w_{us}, w_u\}$
 d. A quit smoking: $I(u \wedge \neg s) = \{w_u\}$

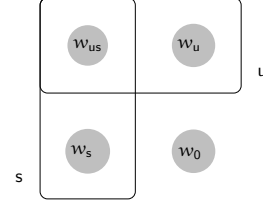


Fig. 1. Model universe

2.2 Global and local contexts

Assume that the input global context is $C_i = D_w$, and that the local context is $C_L = \{w_u\}$, the set of worlds in which the embedded content is true (Figure 2). Unlike in approaches where local contexts are input domains against which presupposition satisfaction is checked (e.g., Heim 1983, 1992; Schlenker 2009), the present account treats presuppositions as not-at-issue entailments rather than conditions on the input context. Accordingly, local contexts are not evaluation domains, but simply sets of worlds that store embedded content.

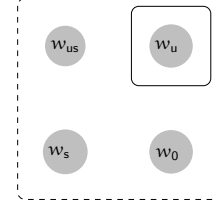


Fig. 2. Input global context $C_i = \{w_{us}, w_s, w_u, w_0\}$ (dashed) and local context in the scope of the modal $C_L = \{w_u\}$ (solid).

2.3 QUD-partitions

The analysis derives projection for entailments that are not at issue relative to the QUD (following Abrusán 2011; Simons et al. 2017; Tonhauser et al. 2018). To derive the contrast between (2) and (3), we assume that the initial questions in the two discourse invoke different question partitions of the context set. In particular, the question in (2), whether Anna smokes, invokes a partition of the context set into worlds where Anna smokes and worlds where she does not, illustrated in Figure 3 (left). In contrast, the question in (3) invokes a partition of the context set into worlds where Anna quit smoking and worlds where did not, shown in Figure 3 (right).

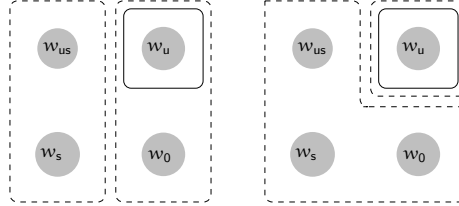


Fig. 3. Local context in the scope of the modal $C_L = \{w_u\}$ (solid), and QUD-partitions over the input global context (dashed) for ?s (2) (left) and $?(u \wedge \neg s)$ (3) (right).

2.4 Similarity and distance from the local context

For concreteness, we formalize similarity between sets of worlds via a distance metric between single worlds w, w' that counts the atomic propositions (wt constants) assigning distinct truth values to w, w' (8).

$$(8) \quad d_w(w') := |\lambda p \in \text{Const}_{wt}. p(w) \neq p(w')|$$

Using a world w as reference point, we can define an equivalence relation (9b), which holds of two worlds w', w'' , iff they are equidistant from w .

$$(9) \quad \sim_w = \lambda w'. \lambda w''. d_w(w') = d_w(w'')$$

For the world in our local context, w_u , the equivalence classes on D_w (shown in Figure 4) are: $\{w_u\}$, $\{w_0, w_{us}\}$, and $\{w_s\}$.

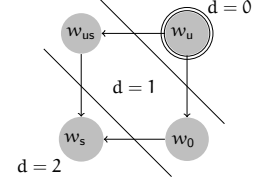


Fig. 4. w_u -similarity classes on D_w

2.5 Similarity-based pruning of QUD-alternatives

The function in (10) selects the subset of p that closest to a reference world w :

$$(10) \quad \text{sim}_{w_u}(p) := \{w' \in p \mid \forall w'' \in p (d_{w_u}(w') \leq d_{w_u}(w''))\}$$

For example, if p contains w_u , then $\text{sim}_{w_u}(p) = \{w_u\}$, selecting the equivalence-class with $d_{w_u} = 0$. If $p = \{w_{us}, w_s, w_0\}$, then $\text{sim}_{w_u}(p) = \{w_{us}, w_0\}$, selecting the equivalence-class with $d_{w_u} = 1$. In general, $\text{sim}_{w_u}(p)$ selects the worlds in p that are in the equivalence class with the lowest distance from w_u .

This notion of similarity is lifted to a relation between context sets in (11), which holds of two sets of worlds p, q , iff for every world in p , there is a world in q such that the world in p is among the most similar worlds to it in p .

$$(11) \quad \llbracket \text{Sim}(p, q) \rrbracket^{M, g} = 1, \text{ iff } \forall w \in p (\exists w' \in q (w \in \text{sim}_{w'}(p)))$$

The proposed analysis uses (11) in context updates to prune QUD-alternatives by retaining, within each alternative, only those worlds that are most similar to the local context. When evaluating the modal assertion of *Maybe Anna quit smoking*, this mechanism selects the worlds in each QUD-alternative that lie in the minimal distance class for at least some world in the local context $C_L = \{w_u\}$. Figure 5 illustrates the effect of this pruning step on the QUD-partitions for (2) and (3).

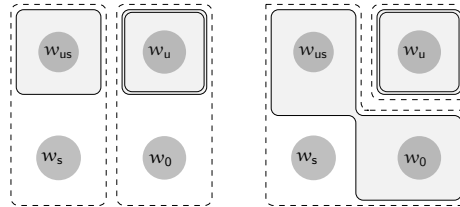


Fig. 5. QUD-partitions over the input global context (dashed) for ?s (2) (left) and ?(u ∧ ¬s) (3) (right), with pruned subsets after applying similarity (shaded).

The assumption of similarity removes from each QUD-alternative, those worlds that are less similar to the local context. In (2) (Fig. 5, left), both resulting alternatives entail the projective inference u . This derives projection, because both QUD-partitions of the context set contain only worlds in which u is true. In contrast, in (3) (Fig. 5, right), the resulting alternatives still contain worlds in which u is false, and thus no projection arises. This approach therefore derives projection as a consequence of cross-alternative invariance and connects to Abusch’s analysis of presupposition triggering from alternative sets (like for focus and questions). Under this proposal, projection can further be understood as a form of global accommodation, based on the default assumptions of similarity between context sets, unless explicitly contradicted.

3 Propositional CDRT with partition

The analysis employs a propositional version of compositional DRT (CDRT, based on Muskens 1996; Brasoveanu 2006, 2010), using propositional discourse referents (drefs) to model local and global context sets (following Stone 1999; Stone and Hardt 1999; Brasoveanu 2006, 2010; AnderBois et al. 2010, 2015; Hofmann 2019, 2022, 2025). Following AnderBois et al. 2010, 2015, the analysis also integrates components of the Table model of Farkas and Bruce 2010 the dynamic system with propositional drefs.

The main innovations proposed here are (i) the use of disjunctive conditions on propositional drefs to introduce question alternatives as alternative values for propositional drefs, and (ii) using a similarity relation over propositional drefs representing local contexts and QUD-alternatives to implement a QUD-sensitive projection mechanism.

3.1 The basic mechanism

Discourse states I, J are modeled as sets of variable assignments, and utterance meanings are *updates* to the current discourse state. The system uses two kinds of updates: Variable updates introduce new drefs and updates with conditions eliminate assignments from the discourse state that do not satisfy the condition. This system uses only propositional drefs $\phi_1, \phi_2, \dots$, which are functions from assignments to sets of possible worlds. Accordingly, for an assignment i , $\phi(i)$ is the proposition stored in ϕ at i .

Variable update given a discourse state I that stores a set of assignments (i.e., a plural discourse state as in Brasoveanu 2006, 2010) applies the singular update distributively to each assignment, as defined in (12).

- $$(12) \quad \begin{array}{ll} \text{a.} & [\phi] := \lambda I. \lambda J. \forall i \in I (\exists j \in J (i[\phi]j)) \wedge \forall j \in J (\exists i \in I (i[\phi]j)) \\ \text{b.} & \text{For assignments } i, j, \text{ dref } \phi: \\ & \llbracket i[\phi]j \rrbracket^{M, g} = 1 \text{ iff for any dref } \phi' \text{ s.t. } \phi' \neq \phi: \phi'(i) = \phi'(j) \end{array}$$

(12b) states that the relation $i[\phi]j$ holds of two singleton assignments iff they differ at most in the value assigned to ϕ . Since ϕ itself is unconstrained, it may be assigned any set of worlds in the output assignment j . (12a) lifts this relation to plural discourse states by applying it relation distributively to the input state I to yield the output J , where each assignment $j \in J$ is derived from some input assignment $i \in I$, and every input assignment $i \in I$ has at least one output assignment $j \in J$.

In our model, introducing a propositional dref ϕ randomly assigns possible values to ϕ . Since propositional values are subsets of U , an update with $[\phi]$ produces one assignment for each possible value of ϕ , as shown in Figure 6.

Updates with conditions require each output assignment to satisfy a stated condition. For example, the condition associated with *Anna smokes*, given in (13), requires each output assignment j to map ϕ to a proposition that entails s . The update therefore eliminates all assignments that do not satisfy this requirement, also illustrated in Figure 6.

$$(13) \quad [\phi \subseteq s] := \lambda I. \lambda J. J \subseteq I \wedge \forall j \in J (\phi(j) \subseteq s)$$

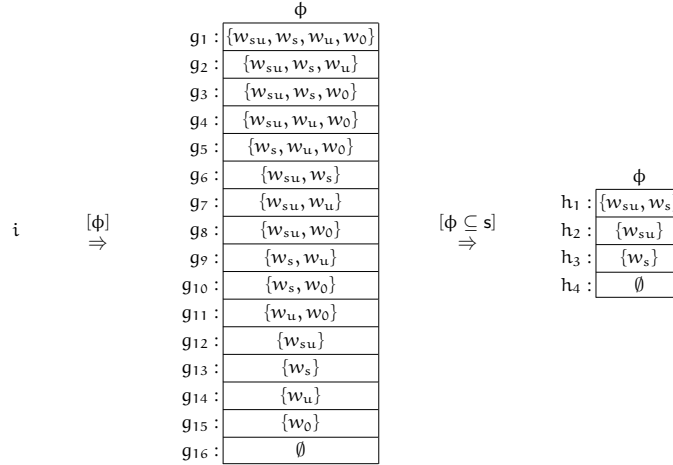


Fig. 6. Result of a simple variable update

3.2 Modeling context sets

To implement components of the Table model of Farkas and Bruce 2010, I assume, following AnderBois et al. 2010, 2015, that discourse states store designated propositional drefs for context-level discourse components. The dref ϕ_C stores the current context set, and ϕ_{PS} stores a projected set of possible future common grounds. In addition, we propose a designated dref ϕ_{QUD} for QUD-partitions of the context set.

An *initial discourse state* devoid of any epistemic information is one where the context set, QUD, and projected set are still compatible with all worlds in D_w . In our model, this is the singleton discourse state g , where $\phi_C(g) = \phi_{QUD}(g) = \phi_{PS}(g) = \{w_{su}, w_s, w_u, w_0\}$, as illustrated in Fig. 7.

ϕ_C	ϕ_{QUD}	ϕ_{PS}
$\{w_{su}, w_s, w_u, w_0\}$	$\{w_{su}, w_s, w_u, w_0\}$	$\{w_{su}, w_s, w_u, w_0\}$

Fig. 7. Initial discourse state

Throughout, the relationship between these discourse components is constrained as follows. Each QUD-alternative is a subset of the context set, and each projected set is a subset of a QUD-alternative. That is, for any assignment i , $\phi_{QUD}(i) \subseteq \phi_C(i)$ and $\phi_{PS}(i) \subseteq \phi_{QUD}(i)$.

3.3 Atomic sentence meanings

I assume, following Wittgenstein (1953); Stenius (1967) that the compositional contribution of an atomic sentence is its propositional content. This content can combine with speech-act operators, such as the polar question operator (Section 3.4) and declarative sentential mood (Section 3.5), or with propositional operators, such as the epistemic possibility modal *maybe* (Section 4).

In this propositional dynamic system, sentence meanings are dynamic propositions evaluated relative to an intensional point of evaluation ϕ . Since points of evaluation are sets of worlds, and since sub-clausal compositionality is not modeled here, a dynamic proposition, like those in (14), takes a propositional dref ϕ as its argument and simply requires its content to entail the proposition characterized by its static translation in the model.

- (14) a. *Anna smokes* $\rightsquigarrow \lambda\phi_w. [\phi \subseteq s]$
 b. *Anna quit smoking* $\rightsquigarrow \lambda\phi_w. [\phi \subseteq (u \wedge \neg s)]$

3.4 Asking a question

The question whether Anna smokes is associated with the update in (15).¹

- (15) *Does Anna smoke?*, i.e., $?(Anna\ smokes) \rightsquigarrow$
 a. $[\phi_1]; \mathbf{max}_{\phi_1}([\phi_1 \subseteq s]);$
 b. $[\phi_{QUD}]; \mathbf{max}_{\phi_{QUD}}([\phi_{QUD} \subseteq \phi_C, \phi_{QUD} \subseteq \phi_1 \vee \text{not}(\phi_{QUD}, \phi_1)])$
 c. $[\phi_{PS}]; [\phi_{PS} = \phi_{QUD}]$

¹ The compositional contribution of the question operator can be stated as:
 $? \rightsquigarrow \lambda P_{wt}. [\phi]; \mathbf{max}_{\phi}(P(\phi));$
 $[\phi_{QUD}]; \mathbf{max}_{\phi_{QUD}}([\phi_{QUD} \subseteq \phi_C, \phi_{QUD} \subseteq \phi \vee \text{not}(\phi_{QUD}, \phi)])$
 $[\phi_{PS}]; [\phi_{PS} = \phi_{QUD}]$

(15) makes three contributions, which are explicated below:

The matrix dref and maximization: The first part of the update, (15a), introduces a propositional dref ϕ_1 for the main clause content, and requires ϕ_1 to map to the largest proposition where that content holds. In this case, ϕ_1 is the largest set of worlds where Anna smokes $\{w_{su}, w_s\}$.

More generally, propositional drefs are introduced throughout as maximal drefs satisfying a given condition (see, e.g., Hofmann 2025, pp. 28–30). The maximization operator is defined in (16).

$$(16) \quad \mathbf{max}_\phi(D) :=$$

- a. $\lambda I_{st}. \lambda J_{st}. D(I)(J) \wedge$
- b. $\forall j_s \in J (\forall K_{st} (D(I)(K) \rightarrow \neg \exists k \in K (\phi(j) \subsetneq \phi(k))))$

(16) states that J is an update of I with D (16a), and that no output value of ϕ in J is strictly smaller than a possible value of ϕ in any alternative output state K obtained by updating I with D (16b). This is a weak maximization condition: it permits values of propositional drefs as long as they are not strictly smaller than another possible value.

This is reflected in Figure 8: after the question update, every output assignment maps ϕ_1 to $\{w_{su}, w_s\}$, the largest set of worlds where Anna smokes.

$$g : \begin{array}{|c|c|c|} \hline \phi_C & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_{lu}, w_0\} & \{w_{su}, w_s, w_{lu}, w_0\} & \{w_{su}, w_s, w_{lu}, w_0\} \\ \hline \end{array} \Rightarrow \begin{array}{l} h_1 : \begin{array}{|c|c|c|c|} \hline \phi_C & \phi_1 & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_{lu}, w_0\} & \{w_{su}, w_s\} & \{w_{su}, w_s\} & \{w_{su}, w_s\} \\ \hline \end{array} \\ h_2 : \begin{array}{|c|c|c|c|} \hline \phi_C & \phi_1 & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_{lu}, w_0\} & \{w_{su}, w_s\} & \{w_{lu}, w_0\} & \{w_{lu}, w_0\} \\ \hline \end{array} \end{array}$$

Fig. 8. Update with ?s for the input context $\{g\}$, where $g(\phi_C) = g(\phi_{QUD}) = g(\phi_{PS}) = U$

Introducing question alternatives: The second part of the update, (16b), introduces a QUD-partition of the context set and stores the QUD-alternatives as possible values of ϕ_{QUD} . This is done by reintroducing ϕ_{QUD} , allowing it to take multiple possible values. These values are constrained to be subsets of ϕ_C that either entail ϕ_1 or are disjoint from ϕ_1 .

The alternatives are formed using the disjunctive condition (17a), which requires each output to satisfy one condition or the other, similarly to program disjunction in DPL (Groenendijk and Stokhof 1990). The negation relation over propositional drefs is given in (17b): which requires that the values of the two drefs are disjoint (based on Stone 1999; Stone and Hardt 1999; AnderBois et al. 2015; Hofmann 2019, 2022, 2025).

$$(17) \quad \begin{array}{ll} \text{a. Disjunctive condition to introduce alternatives:} \\ \phi_{QUD} \subseteq \phi_1 \vee \text{not}(\phi_{QUD}, \phi_1) := \lambda j. \phi_{QUD}(j) \subseteq \phi_1(j) \vee \text{not}(\phi_{QUD}, \phi_1)(j) \\ \text{b. Negation as relation over propositional drefs:} \\ \text{not}(\phi_{QUD}, \phi_1) := \lambda j. \phi_{QUD}(j) \cap \phi_1(j) = \emptyset \end{array}$$

As a result, an update with (15b) reintroduces the dref storing the QUD ϕ_{QUD} and stores maximal subsets of ϕ_{C} that either entail ϕ_1 or are disjoint from ϕ_1 . Given the initial singleton discourse state $\{g\}$, where $g(\phi_{\text{C}}) = g(\phi_{\text{QUD}}) = g(\phi_{\text{PS}}) = \mathbb{U}$, the question *Does Anna smoke?* (15) yields the output state $\{h_1, h_2\}$ shown in Figure 8, above.

Updating the projected set. The final step updates ϕ_{PS} , the dref storing the projected set. As in the Table model (Farkas and Bruce 2010), the question alternatives are also added to the projected set. Here, this is done by (15c), which sets ϕ_{PS} equal to ϕ_{QUD} . Each QUD-alternative therefore corresponds to a possible future common ground, which is also shown in Figure 8, above.

3.5 Assertion

Following Farkas and Bruce (2010) and AnderBois et al. (2010, 2015), assertions are proposals to update the context set ϕ_{C} , but directly update only the projected set ϕ_{PS} . Like a question, an assertion first introduces a dref for the main clause content.² It then updates ϕ_{PS} so that only QUD-alternatives entailed by the asserted content remain possible future common grounds.

- (18) *Anna smokes.*, i.e., $\text{ASSERT}(\text{Anna smokes}) \rightsquigarrow$
 a. $[\phi_2]; \mathbf{max}_{\phi_2}([\phi_2 \subseteq s]);$
 b. $[\phi_{\text{PS}}]; \mathbf{max}_{\phi_{\text{PS}}}([\phi_{\text{PS}} \subseteq \phi_{\text{QUD}}, \phi_{\text{PS}} \subseteq \phi_2])$

(18) introduces ϕ_2 as a dref for the largest set of worlds where Anna smokes, and then reintroduces ϕ_{PS} as the largest subset of the one of the QUD-alternatives and the asserted main clause content ϕ_2 . If *Anna smokes* is asserted after the question in (15), the result is the discourse state in Figure 9. In i_1 , the relevant QUD-alternative is already a subset of ϕ_2 , so $\phi_{\text{PS}} = \{w_{su}, w_s\}$. In i_2 , the QUD-alternative is incompatible with ϕ_2 , so $\phi_{\text{PS}} = \emptyset$. Consequently, only QUD-alternatives that support the assertion remain possible future common grounds.

	ϕ_{C}	ϕ_1	ϕ_{QUD}	ϕ_{PS}		ϕ_{C}	ϕ_1	ϕ_{QUD}	ϕ_2	ϕ_{PS}
$h_1 :$	$\{w_{su}, w_s, w_{uu}, w_0\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s\}$	$\Rightarrow$	$i_1 :$	$\{w_{su}, w_s, w_{uu}, w_0\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s\}$
$h_2 :$	$\{w_{su}, w_s, w_{uu}, w_0\}$	$\{w_{su}, w_s\}$	$\{w_{uu}, w_0\}$	$\{w_{uu}, w_0\}$		$i_2 :$	$\{w_{su}, w_s, w_{uu}, w_0\}$	$\{w_{su}, w_s\}$	$\{w_{uu}, w_0\}$	$\emptyset$

Fig. 9. Update when asserting s after asking whether $?s$.

Accepting the proposal. If the projected set ϕ_{PS} has only one non-empty value in the discourse state, the assertive proposal can be accepted by setting: $[\phi_{\text{C}} = \phi_{\text{PS}}]$. Since the main context set should never be empty in a consistent discourse, this eliminates the QUD-answer rejected by the assertion, and the resulting discourse state will be as given in Figure 10.

² The compositional contribution of the assertion operator can be stated as:
 $\text{ASSERT} \rightsquigarrow \lambda \mathcal{P}_{\text{wt.}}. [\phi]; \mathbf{max}_{\phi}(\mathcal{P}(\phi)); [\phi_{\text{PS}}]; \mathbf{max}_{\phi_{\text{PS}}}([\phi_{\text{PS}} \subseteq \phi_{\text{QUD}}, \phi_{\text{PS}} \subseteq \phi])$

$$j_1 : \begin{array}{|c|c|c|c|c|} \hline \phi_1 & \phi_{\text{QUD}} & \phi_2 & \phi_{\text{PS}} & \phi_c \\ \hline \{w_{su}, w_s\} & \{w_{su}, w_s\} & \{w_{su}, w_s\} & \{w_{su}, w_s\} & \{w_{su}, w_s\} \\ \hline \end{array}$$

Fig. 10. Discourse state after asking whether $?(s)$, asserting and accepting $\text{ASSERT}(s)$.

The machinery introduced in this section provides the basic dynamic treatment of context sets, QUD-alternatives, and projected future common grounds. The next section extends this system to embedded propositional content by defining the updates contributed by epistemic possibility modals. This will allow us to state the projection mechanism formally and derive the QUD-sensitive contrast introduced above.

4 The main contrast

This section applies the dynamic system and similarity-based projection mechanism developed above to the contrast between (2) and (3), repeated here:

- (2) a. Q: *Does Anna smoke?* (?s)
 b. A: *Maybe she quit smoking.* $(\Diamond(u \wedge \neg s))$
 c. Suggests answer: Maybe A doesn't smoke $(\Diamond \neg s)$
 d. Projects “Anna used to smoke” (u)
- (3) a. Q: *Why is Anna nervously chewing her pencil?* $(?(u \wedge \neg s))$
 b. A: *Maybe she quit smoking.* $(\Diamond(u \wedge \neg s))$
 c. Suggests answer: Maybe A quit smoking $(\Diamond(u \wedge \neg s))$
 (and that's why A is chewing her pencil)
 d. No projection

The goal is to derive that the inference that Anna used to smoke projects when the QUD is $?s$, but not when the QUD is $?(u \wedge \neg s)$. Section 4.1 defines the contribution of propositional operators, focusing on the epistemic possibility modal *maybe*, and shows how similarity-based pruning yields projection in the $?s$ case. Section 4.2 then turns to the alternative QUD $?(u \wedge \neg s)$ and shows that the same mechanism correctly predicts the absence of projection.

4.1 Propositional operators and projection from under *maybe*

Maybe. The update associated with an assertion of *Maybe Anna quit smoking* is represented as in (19).³

³ The compositional contribution associated with *not* and *maybe* can be given as:
 $\text{NOT} \rightsquigarrow \lambda \mathcal{P}_{\text{wt}}. \lambda \phi_w. [\phi'; \mathbf{max}(P(\phi')); [\text{not}(\phi, \phi')];$
 $\quad \quad \quad [\phi_{\text{QUD}}]; \mathbf{max}_{\phi_{\text{QUD}}}([\phi_{\text{QUD}} \subseteq \phi_{\text{PS}}, \text{Sim}(\phi_{\text{QUD}}, \phi')])$
 $\text{MAYBE} \rightsquigarrow \lambda \mathcal{P}_{\text{wt}}. \lambda \phi_w. [\phi'; \mathbf{max}(P(\phi')); [\text{might}(\phi, \phi')];$
 $\quad \quad \quad [\phi_{\text{QUD}}]; \mathbf{max}_{\phi_{\text{QUD}}}([\phi_{\text{QUD}} \subseteq \phi_{\text{PS}}, \text{Sim}(\phi_{\text{QUD}}, \phi')])$

- (19) *Maybe Anna quit smoking.* i.e., $\text{ASSERT}(\text{MAYBE } \textit{Anna quit smoking}) \rightsquigarrow$
- a. $[\phi_2]; \mathbf{max}_{\phi_2}([\phi_3]; \mathbf{max}_{\phi_3}([\phi_3 \subseteq s]; [\text{might}(\phi_2, \phi_3)]]);$
 - b. $[\phi_{\text{QUD}}]; \mathbf{max}_{\phi_{\text{QUD}}}([\phi_{\text{QUD}} \subseteq \phi_{\text{PS}}, \text{Sim}(\phi_{\text{QUD}}, \phi')])$
 - c. $[\phi_{\text{PS}}]; \mathbf{max}_{\phi_{\text{PS}}}([\phi_{\text{PS}} \subseteq \phi_{\text{QUD}}, \phi_{\text{PS}} \subseteq \phi_2])$

(19) makes three contributions, which are explicated below.

Introducing drefs for the clausal contents. As shown in Figure 11, below, (19a) introduces a propositional dref ϕ_3 for the prejacent (Stone 1999; Stone and Hardt 1999; Brasoveanu 2006; AnderBois et al. 2015; Hofmann 2022), in this case ϕ_3 is the largest set of worlds in which Anna quit smoking, namely $\{w_u\}$.

(19a) also introduces a dref ϕ_2 for the main clause. Here, ϕ_2 is the largest set of worlds where ϕ_3 might be the case; in this case, this is the set $\{w_{su}, w_s, w_u, w_0\}$, which is derived as follows: First, assuming a Kratzer-style semantics for the modal (following Kratzer 1977, 1981; Stone 1999; Brasoveanu 2006, 2010), we define the condition contributing the semantics of the modal as (20).

$$(20) \quad \text{might}(\phi_2, \phi_3) := \lambda i. (\forall w \in \phi_2(i) (\mathbf{R}(w) \cap \phi_3(i) \neq \emptyset))$$

Second, for concreteness and simplicity of exposition, we assume that the epistemic accessibility relation $\mathbf{R}$ is a *state-based accessibility relation* in ϕ_2 (and our model under consideration), in the sense of Aloni 2022, p. 21:

- (21) A modal accessibility relation $\mathbf{R}$ is state-based in a set of worlds (state) s and model $\mathbf{M}$ iff for all worlds $w \in s : \mathbf{R}(w) = s$.

On this basis, the condition in (20) will be equivalent to that in (22), and therefore simply require that ϕ_2 has a non-empty intersection with ϕ_3 .

$$(22) \quad \text{might}(\phi_2, \phi_3) := \lambda i. (\phi_2(i) \cap \phi_3(i) \neq \emptyset)$$

Since this condition is evaluated in the scope of the maximization operator over ϕ_2 , we pick $\phi_2 = \{w_{su}, w_s, w_u, w_0\}$ as the largest set with a non-empty intersection with ϕ_3 , i.e., no worlds are excluded.

While this dynamic semantics for epistemic possibility modals does not reflect all aspects of their behavior (such as test-like effects, see Veltman 1996; or suggesting an answer to the QUD, e.g., Brumwell 2009), it does reflect that the assertion does not eliminate any epistemic possibilities from the context set, or answers to the QUD. This suffices for present purposes and allows us to isolate the effect of the similarity-based projection mechanism.

Similarity-based projection. The projection of *Anna used to smoke* is derived by the update in (22b), which reintroduces the dref storing the QUD-alternatives ϕ_{QUD} to store for each assignment in the context, the largest subset of ϕ_{QUD} that is most similar to the local context $\phi_3 = \{w_u\}$. The condition imposing similarity is defined in (23), based on the considerations spelled out in Section 2.

$$(23) \quad \text{Sim}(\phi, \phi') := \lambda i. (\forall w \in \phi(i) (\exists w' \in \phi'(i) (w \in \text{sim}_{w'}(\phi(i))))$$

Accordingly, (23b) eliminates all worlds $w \in \phi_{\text{QUD}}$, where another world in the partition is more similar to $\{w_u\}$ than w .

As shown in Figure 11, for $\phi_{\text{QUD}}(h_1) = \{w_{su}, w_s\}$, where s is true ($\{w_{su}, w_s\}$), choosing the most similar worlds to $\{w_u\}$ results in the set $\phi_{\text{QUD}}(i_1) = \{w_{su}\}$. For the other alternative $\phi_{\text{QUD}}(h_2) = \{w_u, w_0\}$, where s is false ($\{w_u, w_0\}$), choosing the most similar worlds to $\{w_u\}$ results in the set $\phi_{\text{QUD}}(i_2) = \{w_u\}$. As a result, all possible answers to the QUD entail u , deriving projection.

	ϕ_C	ϕ_1	ϕ_{QUD}	ϕ_{PS}	
h_1	$\{w_{su}, w_s, w_u, w_0\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s\}$	$\Rightarrow$
h_2	$\{w_{su}, w_s, w_u, w_0\}$	$\{w_{su}, w_s\}$	$\{w_u, w_0\}$	$\{w_u, w_0\}$	

	ϕ_C	ϕ_1	ϕ_2	ϕ_3	ϕ_{QUD}	ϕ_{PS}
i_1	$\{w_{su}, w_s, w_u, w_0\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s, w_u, w_0\}$	$\{w_u\}$	$\{w_{su}\}$	$\{w_{su}\}$
i_2	$\{w_{su}, w_s, w_u, w_0\}$	$\{w_{su}, w_s\}$	$\{w_{su}, w_s, w_u, w_0\}$	$\{w_u\}$	$\{w_u\}$	$\{w_u\}$

Fig. 11. Update when asserting $\Diamond(u \wedge \neg s)$ after asking whether $?s$.

Assertive update of the projected set. Finally, the effect of assertion (23c) updates ϕ_{PS} to contain only answers to the QUD entailed by the assertion. Here, the asserted content ϕ_2 does not restrict the value, but the projected set is narrowed to inheriting the similarity-based pruning of the QUD-alternatives.

Results of the update. After updating with the condition that QUD-alternatives must be similar to the local context ϕ_3 , all QUD-alternatives entail the projective inference u . As a result, all projected future common grounds are restricted to those in which Anna used to smoke, no matter whether the QUD is answered affirmatively (continuing with i_1) or negatively (i_2). Projection thus follows from cross-alternative invariance, derived from the similarity-based projection mechanism.

This derives projection as a form of global accommodation based on a default assumptions of similarity between context sets. A further prediction is that the update excludes worlds in which Anna has just started smoking, excluding the world w_s . This aligns with intuitive judgments, although further research is needed to systematically assess this prediction.

4.2 No projection with $?(u \wedge \neg s)$

The question. We now consider the case in (3), where the QUD is whether there is a change from smoking to non-smoking which could explain Anna's chewing her pencil. Given the initial input $\{g\}$, an update with the question whether Anna quit smoking $?(u \wedge \neg s)$ (24) yields the partition in Figure 12, separating worlds where $u \wedge \neg s$ holds from those where it does not.

- (24) *Did Anna quit smoking*, i.e., $?(Anna \text{ quit smoking}) \rightsquigarrow$
- a. $[\phi_1]; \mathbf{max}_{\phi_1}([\phi_1 \subseteq (u \wedge \neg s)]);$
 - b. $[\phi_{QUD}]; \mathbf{max}_{\phi_{QUD}}([\phi_{QUD} \subseteq \phi_C, \phi_{QUD} \subseteq \phi_1 \vee \text{not}(\phi_{QUD}, \phi_1)])$
 - c. $[\phi_{PS}]; [\phi_{PS} = \phi_{QUD}]$

$$g : \begin{array}{|c|c|c|} \hline \phi_C & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_u, w_0\} & \{w_{su}, w_s, w_u, w_0\} & \{w_{su}, w_s, w_u, w_0\} \\ \hline \end{array} \Rightarrow \begin{array}{l} h_1 : \begin{array}{|c|c|c|c|} \hline \phi_C & \phi_1 & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_u, w_0\} & \{w_u\} & \{w_u\} & \{w_u\} \\ \hline \end{array} \\ h_2 : \begin{array}{|c|c|c|c|} \hline \phi_C & \phi_1 & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_u, w_0\} & \{w_u\} & \{w_{su}, w_s, w_0\} & \{w_{su}, w_s, w_0\} \\ \hline \end{array} \end{array}$$

Fig. 12. Update with $?(u \wedge \neg s)$ for the input context $\{g\}$, where $g(\phi_C) = g(\phi_{QUD}) = g(\phi_{PS}) = U$

The modal update. Applying the modal update in (19), similarity to the local context ϕ_3 is now evaluated relative to a different QUD-partitioning of the context set, which results in a different outcome of the similarity-based pruning. The partitioning of the context set is now based on whether $u \wedge \neg s$ is true or false, so the two QUD-alternatives are $\{w_u\}$ and $\{w_{su}, w_s, w_0\}$.

- (19) *Maybe Anna quit smoking*. i.e., $\text{ASSERT}(\text{MAYBE } Anna \text{ quit smoking}) \rightsquigarrow$
- a. $[\phi_2]; \mathbf{max}_{\phi_2}([\phi_3]; \mathbf{max}_{\phi_3}([\phi_3 \subseteq s]); [\text{might}(\phi_2, \phi_3)]);$
 - b. $[\phi_{QUD}]; \mathbf{max}_{\phi_{QUD}}([\phi_{QUD} \subseteq \phi_{PS}, \text{Sim}(\phi_{QUD}, \phi')])$
 - c. $[\phi_{PS}]; \mathbf{max}_{\phi_{PS}}([\phi_{PS} \subseteq \phi_{QUD}, \phi_{PS} \subseteq \phi_2])$

As shown in Figure 13, for $\phi_{QUD}(h_1) = \{w_u\}$, where $u \wedge \neg s$ is true, choosing the most similar worlds to $\{w_u\}$ results in no changes. For the alternative $h_2(\phi_{QUD}) = \{w_{su}, w_s, w_0\}$, assuming similarity relative to $\{w_u\}$ yields $i_2(\phi_{QUD}) = \{w_{su}, w_0\}$, which does not entail u .

$$\begin{array}{l} h_1 : \begin{array}{|c|c|c|} \hline \phi_C & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_u, w_0\} & \{w_u\} & \{w_u\} \\ \hline \end{array} \\ h_2 : \begin{array}{|c|c|c|} \hline \phi_C & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_u, w_0\} & \{w_{su}, w_s, w_0\} & \{w_{su}, w_s, w_0\} \\ \hline \end{array} \end{array} \Rightarrow \begin{array}{l} i_1 : \begin{array}{|c|c|c|c|c|} \hline \phi_C & \phi_1 & \phi_2 & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_u, w_0\} & \{w_{su}, w_s\} & \{w_u\} & \{w_u\} & \emptyset \\ \hline \end{array} \\ i_2 : \begin{array}{|c|c|c|c|c|} \hline \phi_C & \phi_1 & \phi_2 & \phi_{QUD} & \phi_{PS} \\ \hline \{w_{su}, w_s, w_u, w_0\} & \{w_{su}, w_s\} & \{w_u\} & \{w_{su}, w_0\} & \{w_{su}, w_0\} \\ \hline \end{array} \end{array}$$

Fig. 13. Update when asserting $\neg(u \wedge \neg s)$ after asking whether $?(u \wedge \neg s)$.

As a result, the remaining QUD-alternatives do not uniformly entail u , therefore correctly predicting that the inference that Anna used to smoke does not project in this case, as a consequence of the same similarity-based projection mechanism.

5 Discussion and conclusion

The analysis shows that QUD-sensitive projection can be derived compositionally by combining a dynamic semantics with deriving similarity to the local

context for QUD-alternatives. The analysis developed here provides an explicit derivation of a QUD-sensitive projection mechanism within a compositional dynamic framework, building on approaches that link projection to not-at-issueness (e.g., [Simons 2001](#); [Abrusán 2011](#); [Simons et al. 2017](#); [Tonhauser et al. 2018](#); [Roberts and Simons 2024](#); [Degen and Tonhauser 2025](#)).

The key mechanism is similarity-based pruning of QUD-alternatives. After an update that invokes a hypothetical local context, each QUD-alternative is restricted to those worlds that are maximally similar to this local context. An inference projects just in case it is entailed by all surviving QUD-alternatives. The analysis provides an explicit compositional implementation of QUD-sensitive projection, correctly deriving the main contrast: when the QUD is about whether Anna smokes (?s), the similarity-based projection mechanism yields alternatives that uniformly entail that Anna used to smoke (u), deriving projection. When the QUD is about whether Anna quit smoking (?($u \wedge \neg s$)), similarity preserves alternatives that differ with respect to u, and no projection arises.

This mechanism also makes predictions for a QUD targeting the presupposed content u:

- | | | | |
|------|----|--|---------------------------------|
| (25) | a. | Q: <i>Did Anna smoke in the past?</i> | (?u) |
| | b. | A: <i>Maybe she quit.</i> | ($\Diamond(u \wedge \neg s)$) |
| | c. | Suggests answer: Maybe A used to smoke | ($\Diamond u$) |
| | d. | Projects “Anna doesn’t smoke now” | ($\neg s$) |

Here, the QUD renders u at-issue and $\neg s$ not-at-issue. The proposed analysis therefore predicts projection of $\neg s$ via the same similarity-based mechanism. While the answer in (25) is often judged as degraded, given the question, inviting further study, the inference in (25d) does appear to be available, when (25) is accepted. This work provides a straightforward explanation for this inference, as a consequence of the same mechanism that derives the main contrast. More generally, this work provides a proof of concept for systematically deriving projection for not-at-issue entailments.

An important direction for future work concerns empirical patterns where projection does not correlate straightforwardly with at-issueness. [Degen and Tonhauser 2025](#) report that at-issueness correlates negatively with projection for the contents of the complements of many clause-embedding predicates, so that the less at-issue a content is, the more it projects, and vice versa. However, they also report that for some predicates, such as *pretend* and *think*, projection correlates positively with at-issueness ratings, and for a range of clause-embedding predicates, no correlation was found. Within the present framework, additional patterns cases could be modeled by weakening or suspending the default similarity assumption, thereby allowing greater divergence between local and global contexts. Exploring when and how this assumption can be overridden may provide a route to explaining variation in projection behavior across constructions and discourse contexts.

Bibliography

- Abbott, Barbara. 2000. Presuppositions as nonassertions. *Journal of Pragmatics* 32(10):1419–1437.
- Abrusán, Márta. 2011. Predicting the presuppositions of soft triggers. *Linguistics and Philosophy* 34(6):491–535.
- Abusch, Dorit. 2010. Presupposition triggering from alternatives. *Journal of Semantics* 27(1):37–80.
- Aloni, Maria. 2022. Logic and conversation: the case of free choice. *Semantics and Pragmatics* 15:5–1.
- AnderBois, Scott, Adrian Brasoveanu, and Robert Henderson. 2010. Crossing the appositive/at-issue meaning boundary. In *Semantics and linguistic theory*, vol. 20, pages 328–346.
- AnderBois, Scott, Adrian Brasoveanu, and Robert Henderson. 2015. At-issue proposals and appositive impositions in discourse. *Journal of Semantics* 32(1):93–138.
- Brasoveanu, Adrian. 2006. *Structured nominal and modal reference*. Doctoral dissertation, Rutgers University, New Brunswick, NJ.
- Brasoveanu, Adrian. 2010. Decomposing modal quantification. *Journal of Semantics* 27(4):437–527.
- Brumwell, Chris. 2009. A dynamic analysis of epistemic possibility .
- Büring, Daniel. 2003. On D-Trees, Beans, And B-Accents. *Linguistics and Philosophy* 26(5):511–545.
- Degen, Judith and Judith Tonhauser. 2025. Projection inferences: On the relation between prior beliefs, at-issueness, and lexical meaning. *Glossa Psycholinguistics* 4(1).
- Farkas, Donka and Kim Bruce. 2010. On reacting to assertions and polar questions. *Journal of Semantics* 27(1):81–118.
- Geurts, Bart. 1994. *Presupposing*. Doctoral dissertation, Universität Osnabrück.
- Ginzburg, Jonathan. 1996. Interrogatives: Questions, facts and dialogue. *The handbook of contemporary semantic theory* 5(18):359–423.
- Groenendijk, J. and M. Stokhof. 1990. Dynamic Montague Grammar. In L. Kálman and L. Pólos, eds., *Papers from the Second Symposium on Logic and Language*, pages 3–48. Budapest: Akadémiai Kiadó, Budapest.
- Heim, Irene. 1983. On the projection problem for presuppositions. In M. T. Wescoat and M. Barlow, eds., *Second West Coast Conference on Formal Linguistics (WCCFL)*, pages 114–125. Stanford, CA. Reprinted in Portner, Paul and Barbara Partee (eds.). 2002. *Formal semantics: The essential Readings* Linguistics: The Essential Readings, 249260. Hoboken, NJ: Wiley-Blackwell.
- Heim, Irene. 1992. Presupposition projection and the semantics of attitude verbs. *Journal of Semantics* 9(3):183–221.
- Hofmann, Lisa. 2019. The anaphoric potential of indefinites under negation and disjunction. In J. J. Schlöder, D. McHugh, and F. Roelofsen, eds., *22nd Amsterdam Colloquium*, pages 181–190.

- Hofmann, Lisa. 2022. *Anaphora and negation*. Doctoral dissertation, University of California Santa Cruz.
- Hofmann, Lisa. 2025. Anaphoric accessibility with flat update. *Semantics and Pragmatics* 18(3).
- Karttunen, Lauri. 1974. Presupposition and linguistic context. *Theoretical linguistics* 1(1-3):181–194.
- Kratzer, Angelika. 1977. What must and can must and can mean. *Linguistics & Philosophy* 1:337–355.
- Kratzer, Angelika. 1981. The notional category of modality. In H.-J. Eikmeyer and H. Rieser, eds., *Words, worlds and contexts*, New approaches in world semantics, pages 289–323. De Gruyter.
- Lewis, David. 1973. *Counterfactuals*. Cambridge, MA: Harvard University Press.
- Muskens, Reinhard. 1996. Combining Montague Semantics and discourse representation. *Linguistics and Philosophy* 19(2):143–186.
- Qing, Ciyang, Noah D. Goodman, and Daniel Lassiter. 2016. A rational speech-act model of projective content. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 38.
- Roberts, Craige. 2012. Information Structure: Towards an integrated formal theory of pragmatics. *Semantics and Pragmatics* 5:6:1–69.
- Roberts, Craige and Mandy Simons. 2024. Preconditions and projection: Explaining non-anaphoric presupposition. *Linguistics and Philosophy* 47(4):703–748.
- Schlenker, Philippe. 2009. Local contexts. *Semantics and Pragmatics* 2(3):1–78.
- Scontras, Gregory and Judith Tonhauser. 2025. Projection without lexically-specified presupposition: A model for ‘know’. In *Proceedings of Sinn und Bedeutung*, vol. 29, pages 1431–1448.
- Sharvit, Yael. 1998. Individual concepts and attitude reports. In *Semantics and Linguistic Theory*, pages 233–248.
- Simons, Mandy. 2001. On the conversational basis of some presuppositions. *Semantics and Linguistics Theory* 11:431–448. Publisher: Ithaca, NY: CLC Publications.
- Simons, Mandy, David Beaver, Craige Roberts, and Judith Tonhauser. 2017. The best question: Explaining the projection behavior of factives. *Discourse Processes* 54(3):187–206.
- Simons, Mandy, Judith Tonhauser, David Beaver, and Craige Roberts. 2010. What projects and why. In *Semantics and linguistic theory*, vol. 20, pages 309–327.
- Snider, Todd. 2017. *Anaphoric reference to propositions*. Doctoral dissertation, Cornell University, Ithaca, NY.
- Stalnaker, Robert C. 1968. A theory of conditionals. In N. Resher, ed., *Studies in Logical Theory*. Oxford, UK: Blackwell.
- Stenius, Erik. 1967. Mood and language-game. *Synthese* 17(3):254–274.
- Stone, Matthew. 1999. Reference to possible worlds. Tech. Rep. 49, Rutgers Center for Cognitive Science, New Brunswick, NJ.
- Stone, Matthew and Daniel Hardt. 1999. Dynamic discourse referents for tense and modals. In H. Bunt and R. Muskens, eds., *Computing meaning*, vol. 73 of

- Studies in Linguistics and Philosophy*, pages 301–319. Dordrecht, Netherlands: Springer.
- Sudo, Yasutada. 2014. On de re predicates. In *Proceedings of WCCFL*, vol. 31, pages 447–456.
- Tonhauser, Judith, David I. Beaver, and Judith Degen. 2018. How projective is projective content? Gradience in projectivity and at-issueness. *Journal of Semantics* 35(3):495–542.
- Veltman, Frank. 1996. Defaults in update semantics. *Journal of philosophical logic* 25(3):221–261.
- Warstadt, Alex. 2022. Presupposition triggering reflects pragmatic reasoning about utterance utility. In *Proceedings of the Amsterdam Colloquium*, pages 451–458.
- Wittgenstein, Ludwig. 1953. *Philosophische Untersuchungen*. Oxford, UK: Blackwell.